

Assignment Scope: **lectures 1 - 3**

Maximum Grade: 5

Student Name:

Student ID:

Assignment Rules & Regulations:

- Plagiarism, cheating, or any form of academic dishonesty is **strictly prohibited**. All work submitted must be the student's own, and all external sources, such as code snippets, libraries, or academic papers, must be properly cited.
- Requests for extensions must be requested **in advance of the deadline** and should be supported by valid and documented reasons, such as medical or family emergencies. Extensions will be granted at the instructor's discretion.
- If you encounter any issues or require clarifications related to the assignment, promptly communicate with the instructor or teaching assistant. **Do not share your assignment solutions with classmates before the submission deadline.**

Assignment Objectives:

- Topic#1: Data organization, presentation, and summarization.
- Topic#2: Basic concepts in probability theory.

Assignment Exercises:

Exercise 1: A survey was taken of 50 individuals. They were asked how many days per week they ate at a fast-food restaurant. Construct a frequency distribution using 8 classes (0–7). Based on the distribution, how often did most people eat at a fast-food restaurant?

1	3	4	0	4
5	2	2	3	1
2	2	2	2	2
2	2	2	2	3
2	2	5	2	4
2	4	5	2	1
4	1	3	2	2
2	0	7	2	3
2	2	2	5	2
3	3	4	1	3

Exercise 2: For a number of randomly selected college applicants, the following frequency distribution for entrance exam scores was obtained. Construct a histogram, frequency polygon, and ogive for the data.

Class limits	Frequency
90–98	6
99–107	22
108–116	43
117–125	28
126–134	9
Total 108	

Exercise 3: One of the air pollutants that is measured in selected cities is sulfur dioxide. This pollutant occurs when fossil fuels are burned. This pollutant is measured in $\mu\text{g}/\text{m}^3$. The results obtained from a sample of 24 cities are shown in the frequency distributions. One sample was taken recently, and



the other sample of the same cities was taken 5 years ago. Construct a histogram and compare the two distributions.

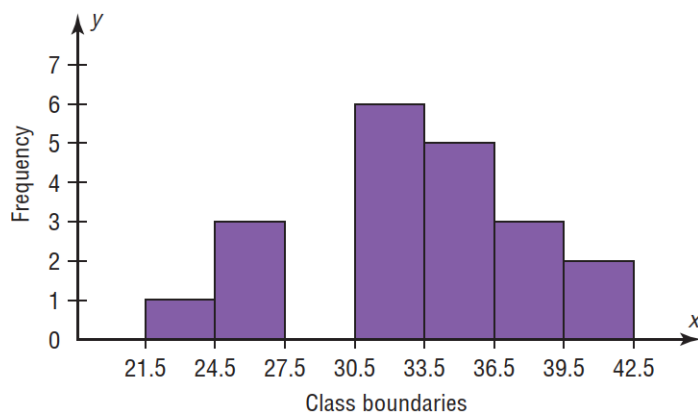
Class limits	Frequency (now)	Frequency (5 years ago)
10–14	6	5
15–19	4	4
20–24	3	2
25–29	2	3
30–34	5	6
35–39	1	2
40–44	2	1
45–49	1	1
	Total 24	Total 24

Exercise 4: The frequency distribution shows the blood glucose levels (in milligrams per deciliter) for 50 patients at a medical facility. Construct a histogram, frequency polygon, and ogive for the data. Comment on the shape of the distribution. What range of glucose levels did most patients fall into?

Class limits	Frequency
60–64	2
65–69	1
70–74	5
75–79	12
80–84	18
85–89	6
90–94	5
95–99	1
	Total 50

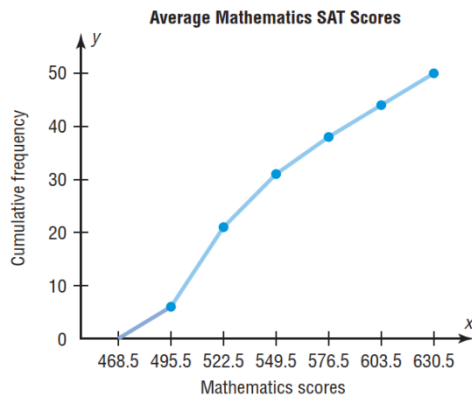
Exercise 5: Using the histogram shown here, do the following.

- Construct a frequency distribution; include class limits, class frequencies, midpoints, and cumulative frequencies.
- Construct a frequency polygon.
- Construct an ogive.



Exercise 6: Shown is an ogive depicting the cumulative frequency of the average mathematics SAT scores by state. Use it to construct a histogram and a frequency polygon.





Exercise 7: Several automobiles were tested for fuel efficiency (in miles per gallon). This frequency distribution was obtained. Find the mean and modal class for the data.

Class boundaries	Frequency
7.5–12.5	3
12.5–17.5	5
17.5–22.5	15
22.5–27.5	5
27.5–32.5	2

Exercise 8: A random sample of a number of states shows the number of specialty coffee shops for a specific company. Find the mean and modal class for the data.

Class boundaries	Frequency
0.5–19.5	12
19.5–38.5	7
38.5–57.5	5
57.5–76.5	3
76.5–95.5	3

Exercise 9: The given frequency distribution represents the data obtained from a sample of a number of copying machine service technicians. The values represent the days between service calls for various copying machines. Find the mean and modal class for the data.

Class boundaries	Frequency
15.5–18.5	14
18.5–21.5	12
21.5–24.5	18
24.5–27.5	10
27.5–30.5	15
30.5–33.5	6

Exercise 10: Find the mean of 10, 20, 30, 40, and 50.

- Add 10 to each value and find the mean.
- Subtract 10 from each value and find the mean.
- Multiply each value by 10 and find the mean.
- Divide each value by 10 and find the mean.
- Make a general statement about each situation.



Exercise 11: The scores from the last 10 years' Bowling Congress Championships are shown below for men and for women. Find the range, variance, and standard deviation for each data set. Which set of data is more variable?

Men 823 837 858 791 812 814 832 862 833 826
 Women 752 764 754 774 771 745 736 816 792 793

Exercise 12: The costs per load (in cents) of 35 laundry detergents tested by a consumer organization are shown here. Find the variance and standard deviation for the data.

Class limits	13 - 19	20 - 26	27 - 33	34 - 40	41 - 47	48 - 54	55 - 61	62 - 68
Frequency	2	7	12	5	6	1	0	2

Exercise 13: The average number of days that construction workers miss per year is 11. The standard deviation is 2.3. The average number of days that factory workers miss per year is 8 with a standard deviation of 1.8. Which class is more variable in terms of days missed?

Exercise 14: The mean of the waiting times in an emergency room is 80.2 minutes with a standard deviation of 10.5 minutes for people who are admitted for additional treatment. The mean waiting time for patients who are discharged after receiving treatment is 120.6 minutes with a standard deviation of 18.3 minutes. Which times are more variable?

Exercise 15: Americans spend an average of 3 hours per day online. If the standard deviation is 32 minutes, find the range in which at least 88.89% of the data will lie.

Exercise 16: The national average for mathematics SATs in 2011 was 514. Suppose that the distribution of scores was approximately bell-shaped and that the standard deviation was approximately 40. Within what boundaries would you expect 68% of the scores to fall? What percentage of scores would be above 594?

Exercise 17: Classify each statement as an example of classical probability, empirical probability, or subjective probability.

- The probability that a person will watch the 6 o'clock evening news is 0.15.
- The probability of winning at a Chuck-a-Luck game is $\frac{5}{36}$.
- The probability that a bus will be in an accident on a specific run is about 6%.
- The probability of getting a royal flush when five cards are selected at random is $\frac{1}{649740}$.

Exercise 18: The table below represents the college degrees awarded in a recent academic year by gender. Choose a degree at random. Find the probability that it is

	Bachelor's	Master's	Doctorate
a. A bachelor's degree			
b. A doctorate or a degree awarded to a woman	Men 573,079	211,381	24,341
c. A doctorate awarded to a woman	Women 775,424	301,264	21,683
d. Not a master's degree			

Exercise 19: The frequency distribution shown here illustrates the number of medical tests



conducted on a number of randomly selected emergency patients. If a patient is selected at random, find these probabilities.

- a. The patient has had exactly 2 tests done.
- b. The patient has had at least 2 tests done.
- c. The patient has had at most 3 tests done.
- d. The patient has had 3 or fewer tests done.
- e. The patient has had 1 or 2 tests done.

Number of tests performed	0	1	2	3	4 or more
Number of patients	12	8	2	3	5

Exercise 20: State which events are independent and which are dependent.

- a. Tossing a coin and drawing a card from a deck.
- b. Drawing a ball from an urn, not replacing it, and then drawing a second ball.
- c. Getting a raise in salary and purchasing a new car.
- d. Driving on ice and having an accident.

Exercise 21: Thirty-five percent of adults who own cell phones use their phones to send and receive text messages. Choose 4 cell phone owners at random. What is the probability that none use their phones for texting?

Exercise 22: Twenty-five percent of all deaths (all ages) are caused by diseases of the heart. Ischemic heart disease accounts for 16.4% of all deaths and heart failure for 2.3%. Choose one death at random. What is the probability that it is from ischemic heart disease given that it was from heart disease? Choose two deaths at random; what is the probability that at least one is from heart disease?

Deadline & Submission: A soft copy of your assignment solution must be submitted latest by Thursday, October 30th, 2025.

